

Dylan Bailes

MECHANICAL ENGINEERING | CONTROLS | EMBEDDED SYSTEMS

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U.S. citizen | Eligible for security clearance

SUMMARY

Mechanical engineering M.S. candidate at UC San Diego (June 2027) with unmanned aerial systems (SUAS) experience at AeroVironment and hands-on work in electromechanical design, controls, embedded firmware, and robotics. Connects modeling and simulation to PCB design, fabrication, integration, and testing.

EDUCATION

M.S. Mechanical Engineering | University of California, San Diego

Expected Jun 2027

Coursework: Optimal Linear Control, Nonlinear Control, Parametric Identification of Systems, Embedded Systems

B.S. Mechanical Engineering | University of California, San Diego

Jun 2026

Controls and Robotics specialization | GPA 3.62/4.00 | Provost Honors | Degree requirements completed in three years

Coursework: Linear Control Design (Kalman Filter, H-Infinity), Intro to Autonomous Vehicles, Dynamics and Control of Aerospace Vehicles, Orbital Mechanics, Advanced Vibrations, Solid Mechanics I & II, Machine Learning Algorithms, Linear Circuits

TECHNICAL SKILLS

Controls and software: Python, C++, MATLAB, Java, ROS2, Linux; Kalman filtering, state-space control, PID, MPC

Embedded and electronics: STM32 (HAL, direct register access), I2C, UART, ESP32, Raspberry Pi, Arduino, KiCad PCB design

Design and analysis: SolidWorks (incl. macros), Fusion 360, ANSYS FEA and Maxwell, GD&T

Fabrication: CNC, manual mill and lathe, rapid prototyping, design for manufacturing

EXPERIENCE

Mechanical Engineering Intern | AeroVironment | Simi Valley, CA

Jun–Aug 2025

- Designed and prototyped a motor-driven SUAS mechanism under tight mass, packaging, and load constraints; completed the prototype and initial design for one subassembly with cross-functional engineering teams.
- Developed physics-based motor and spring models; a 99.5%-confidence GD&T/tolerance analysis showed insufficient spring-force margin at the low end of the tolerance range.
- Built SolidWorks macros and FEA automation that refined meshes near holes and tight corners, fit stress-convergence curves, and estimated asymptotic peak stress, reducing design iteration time.
- Worked with cross-functional teams to advance subassemblies through initial design; tested motor-driven mechanisms against weight and strength requirements.

Mechanical Lead | Robodex FIRST Robotics Competition | Granada Hills, CA

Aug 2019–Jun 2023

- Led mechanical design of two award-winning FRC robots, including the team's first regional competition win in its history; fabricated 300+ parts across CAD (Fusion 360, SolidWorks), CNC, and manual machining.
- Logged 1,000+ hands-on hours, integrated mechanisms with electronics and programming teams, and developed and taught an elementary-school robotics curriculum.

Audio Engineer / Intern | RK Media | Thousand Oaks, CA

Jun 2022–Oct 2024

- Wrote an automated file-sorting tool to streamline the team's media organization workflow.

PROJECTS

Multi-Chamber Camera Bioreactor | UCSD, team of 4, Faculty Affiliate: Dr. Peter Chen

Winter–Spring 2026

- Designed, built, programmed, simulated, and tested the magnetic-field subsystem of a test setup studying stem cell behavior under electric/magnetic fields and fluid flow; measured 4.98 mT at coil center with under 1% variation across the well footprint.
- Simulated electromagnet geometry in ANSYS Maxwell; designed the drive/control electronics, two KiCad PCBs, and STM32 firmware (I2C, UART) with a Raspberry Pi orchestrating operation and an ESP32 used in early prototyping.
- Debugged option-byte settings and used direct register access when HAL failed.

Autonomous Car Racing | UC San Diego, MAE 148

Spring 2026

- Manufactured and integrated a ROS2 race-car platform with camera, LiDAR, and GPS sensing alongside three ECE teammates.
- Coordinated deep-learning perception, Kalman-filter state estimation, and state-space control for robust predictive operation.

Flexible-Shaft Torque Control Testbed | In progress

2026

- Completed mechanical design and controller code for a compliant music-wire torque testbed comparing PID, MPC, and learned MPC; hardware validation pending parts.

Competition Robot | UC San Diego, MAE 3 (Design and Manufacturing Lead)

2023

- Designed a robot with a 3-stage differential lift system; completed 50+ design iterations.